Basics of FE-Analysis
BK10A6400
Lecture 7: Isoparametric description

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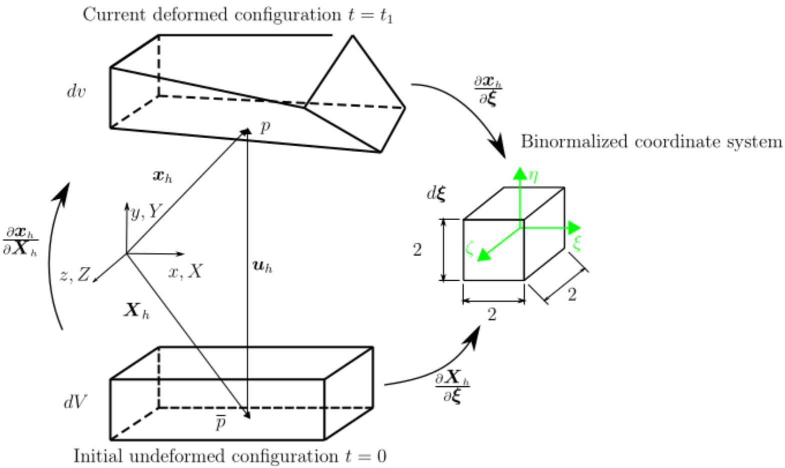
Agenda

- Exercise 3: Plane structure with a hole.
- Bonus task 6: cantilever problem analyzed with four-node plane elements.
- Isoparametric description: what it is and what it means?

Continuum mechanics: Position, displacement and deformations



Position, displacement and deformations

The deformation (and rigid body motion) between the initial and current configurations is described via the deformation gradient as

$$\boldsymbol{F} = \frac{\partial \boldsymbol{x}_h}{\partial \boldsymbol{X}_h} = \frac{\partial \boldsymbol{x}_h}{\partial \boldsymbol{\xi}} \left(\frac{\partial \boldsymbol{X}_h}{\partial \boldsymbol{\xi}} \right)^{-1} = \boldsymbol{j}_e \boldsymbol{J}_e^{-1} \tag{1}$$

The isoparametric transformation between the initial configuration and binormalized (natural) coordinates:

$$\boldsymbol{J}_e = \frac{\partial \boldsymbol{X}_h}{\partial \boldsymbol{\xi}} \tag{2}$$

The isoparametric transformation between the current (deformed) configuration and binormalized (natural) coordinates:

$$\boldsymbol{j}_e = \frac{\partial \boldsymbol{x}_h}{\partial \boldsymbol{\xi}} \tag{3}$$

Isoparametric description

- Previously, interpolation functions are chosen to approximate the primary field variable.
- The exact solution for the displacement field of one element Ω_e is approximated as follows

$$\boldsymbol{u}_{exact} \approx \boldsymbol{u}_h = \sum_{I=1}^{n} N_I \boldsymbol{u}_I \tag{4}$$

- The local shape functions $N_I(x, y, z)$ can be expressed as local reference shape functions $N_I(\xi, \eta, \zeta)$ with local reference coordinates.
- For example, interpolation for the two-dimensional four-node solid (plane) element (Q4, 2412)

$$\begin{aligned} \boldsymbol{u}_{exact} \approx \boldsymbol{u}_h &= \sum_{I=1}^{4} N_I(x, y) \boldsymbol{u}_I = \sum_{I=1}^{4} N_I(\xi, \eta) \boldsymbol{u}_I \\ &= N_1 \boldsymbol{u}_1 + N_2 \boldsymbol{u}_2 + N_3 \boldsymbol{u}_3 + N_4 \boldsymbol{u}_4 = \boldsymbol{N} \boldsymbol{u} \end{aligned} \tag{5}$$

Local reference coordinates, local (physical) coordinates

- Local (physical) element coordinates $\boldsymbol{x} = \{x, y, z\}$ in 3D.
- Local reference (binormalized due to the numerical integration) coordinates $\boldsymbol{\xi} = \{\xi, \eta, \zeta\}$ in 3D

Isoparametric description

$$\boldsymbol{u}_h = \sum_{I=1}^{n} N_I(x,y)\boldsymbol{u}_I = \sum_{I=1}^{n} N_I(\xi,\eta)\boldsymbol{u}_I = \boldsymbol{N}(\xi,\eta)\boldsymbol{u}$$
$$\boldsymbol{X} = \sum_{I=1}^{n} N_I(x,y)\boldsymbol{X}_I = \sum_{I=1}^{n} N_I(\xi,\eta)\boldsymbol{X}_I = \boldsymbol{N}(\xi,\eta)\boldsymbol{X} \tag{6}$$

- $\boldsymbol{u}_h$ is an interpolated displacement field within the element at the current configuration
- $\boldsymbol{u}_I$ is a vector of nodal displacements of a node I at the element's current configuration
- $\boldsymbol{X}_h$ is an interpolated position field within the element at the initial configuration
- $\boldsymbol{X}_I$ is a vector of nodal positions of a node I at the element's initial configuration

Example of isoparametric description: one-dimensional interpolation

$$u_h = \sum_{I=1}^{n} N_I(\xi) u_I = \boldsymbol{N}\boldsymbol{u}$$
$$X_h = \sum_{I=1}^{n} N_I(\xi) X_I = \boldsymbol{N}\boldsymbol{X} \tag{7}$$

where $\boldsymbol{N} = \{N_1, N_2, \ldots, N_I\}$, $\boldsymbol{u} = \{u_1, u_2, \ldots, u_I\}^T$ and $\boldsymbol{X} = \{X_1, X_2, \ldots, X_I\}^T$. The derivatives w.r.t physical coordinates can be obtained (chain rule) as follows:

$$\frac{\partial u_h}{\partial X_h} = \frac{\partial u_h}{\partial \xi}\frac{\partial \xi}{\partial X_h}$$
$$= \frac{\partial u_h}{\partial \xi}\left(\frac{\partial X_h}{\partial \xi}\right)^{-1} = \frac{\partial u_h}{\partial \xi} J_e^{-1} \tag{8}$$

Example of isoparametric description: one-dimensional interpolation

So the abbreviation J_e for a scaling factor between r_0 and ξ can be written with shape functions and global nodal coordinates as follows

$$
\begin{aligned}
J_e &= \frac{\partial X_h}{\partial \xi} \\
&= \sum_{I=1}^{n} \frac{\partial N_I}{\partial \xi} X_I
\end{aligned}
\tag{9}
$$

As can be seen from Eq.(9), the scaling factor J_e depends on shape functions and nodal position coordinates q_0. Scaling factor J_e is needed for the numerical integration of the stiffness matrix and load vector. Mapping (or transformation) J_e is uniquely defined and it keeps its direction if it is always positive

$$
J_e > 0 \tag{10}
$$

Example of transformation between configurations: line element of a two-dimensional plate element

Let's do mapping for a piece of a small infitinesimal line δL at physical (current) configuration.
Transformation from physical coordination to binormalized cooridnate system

$$\delta\boldsymbol{\xi}_h = \frac{\partial \boldsymbol{X}_h}{\partial \boldsymbol{\xi}} \delta \boldsymbol{X}_h \tag{11}$$